

Bibek Poudel

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Experience

Fluidic City Lab & Center for Transportation Research, University of Tennessee

2023 – May 2026

Research Assistant – Reinforcement Learning (RL) and Robotics

- Led RL research on real-world optimization problems in transportation and robotics; developed multi-agent, co-design, and human-in-the-loop RL across simulation and physical hardware; published 7+ papers (4 first-author) across IROS/ICRA/ CHASE/ UbiComp.
- Selected result: developed a search-and-planning algorithm for sequential transit network design; improved solution quality by 29% over end-to-end RL while using 35% fewer training samples.

Department of Computer Science, University of Memphis

2019 – 2023

Research Assistant – Real-time Control and Adversarial Machine Learning

- Applied sample-efficient RL to real-time DC motor position control, built sim-to-hardware policy transfer pipeline; used adversarial ML to assess robustness of traffic-state prediction models and evaluated resilience to black-box attacks.
- Selected result: trained DC-motor steering control policies in under 2 minutes (simulation) and 10 minutes (hardware).

Publications

- **B. Poudel**, S. Swaminathan, W. Li “AlphaTransit: Learning to Design City-scale Transit Routes.” *Under Review*.
- **B. Poudel**, L. Zhu, K. Heaslip, S. Swaminathan, W. Li “DeCoR: Design and Control Co-Optimization for Urban Streets using Reinforcement Learning.” *Under Review*.
- A. Sastry, **B. Poudel**, W. Li “Sparsification of Dynamic Graphs in Robotic Exploration Algorithms.” *Under Review*.
- T. Wu, Y. Wu, **B. Poudel**, S. Meerza, R. Gore, W. Li, Z. Gao, C. Karatas, J. Liu, “VibRun: Real-time Unobtrusive Gait Analysis for Treadmill Running via Footstep Vibrations.” *ACM UbiComp 2025*.
- **B. Poudel**, X. Wang, W. Li, L. Zhu, K. Heaslip, “Joint Pedestrian and Vehicle Traffic Optimization in Urban Environments using Reinforcement Learning.” *IEEE IROS 2025*.
- A. Zahid, **B. Poudel**, D. Scott, J. Scott, S. Crouter, W. Li, S. Swaminathan, “PulseRide: A Robotic Wheelchair for Personalized Exertion Control with Human-in-the-Loop Reinforcement Learning.” *IEEE/ACM CHASE 2025*.
- **B. Poudel**, W. Li, K. Heaslip, “EnduRL: Enhancing Safety, Stability, and Efficiency of Mixed Traffic Under Real-World Perturbations via Reinforcement Learning.” *IEEE IROS 2024*.
- M. Villarreal, **B. Poudel**, J. Pan, W. Li. “Mixed Traffic Control and Coordination from Pixels.” *IEEE ICRA 2024*.

See more in [Google Scholar](#)

Projects

- [Cowrite.io](#)
 - * Created an AI-assisted scientific writing and collaboration tool for researchers (10+ active users).
 - * Used Firebase functions to host a LaTeX compiler and real-time database for seamless multi-user collaboration.
 - * Skills: React, Typescript, Firebase (Authentication, Storage, Functions), Gemini API.
- Documint: [Paper](#) · [GitHub](#) · [HuggingFace](#) (50 downloads/month)
 - * Led a team of 4 researchers to develop a fine-tuning dataset (100k samples) to improve the quality of docstrings generated by Code LLMs. Improved Google’s CodeGemma model by up to 35%.
 - * Skills: LoRA, PyTorch, HuggingFace, Weights & Biases.
- Robustness in Autonomous Steering: Used self-supervised learning to improve adversarial robustness of computer vision models in steering angle prediction under disturbances to camera such as rain, snow, fog, frost, and blur. [PDF](#)
- Defending game playing RL agents against attacks: Developed input feature squeezing as a defense against adversarial attacks on RL agents, enabling them to win all games that would otherwise have been lost. [PDF](#)

Education

Ph.D. in Computer Science

2023 – May 2026

University of Tennessee, Knoxville

M.S. in Computer Science

2019 – 2023

University of Memphis

GPA: 4.0/4.0

Honors & Awards

- Best paper award finalist: “Congestion-Aware Reinforcement Learning” paper in *IEEE CYBER 2024*.
- Best project award winner: BarterBaron in COMP 7012 Software Engineering, U of M.